Lab Manual 1

Practice Commands for This Lab

Command	Purpose	Example (based on sample files)	Effect / Output
`echo`	Print text or variables	`echo "Hello OS Lab"`	Displays `Hello OS Lab`
`cd`	Change directory	`cd ~/os-lab`	Moves into `os-lab`
`ls`	List files	`ls`	`hello.txt copied.txt`
`ls -l`	Long list (permissions, owner, size)	`ls -l`	`-rw-r--r-- 1 farhan users 12 Sep 19 hello.txt`
`ls -a`	Show hidden files too	`ls -a`	`. .. .bashrc hello.txt copied.txt`
`mkdir`	Create a directory	`mkdir ~/new-lab`	Creates `new-lab`
`touch`	Create empty file / update timestamp	`touch ~/os-lab/hello.txt`	Makes `hello.txt` if missing
`cat`	Show file contents	`cat ~/os-lab/hello.txt`	Prints file text
`nano`	Open file in text editor	`nano ~/os-lab/hello.txt`	Edit file inside terminal
`file`	Show file type	`file ~/os-lab/hello.txt`	`ASCII text`
`cp`	Copy files	`cp hello.txt copied.txt`	Creates `copied.txt`
`diff`	Compare two files	`diff hello.txt copied.txt`	Shows line differences
`mv`	Move / rename file	`mv hello.txt ~/new-lab/renamed.txt`	File renamed/moved
`rm`	Delete file	`rm copied.txt`	Removes `copied.txt`
`rm -r`	Delete folder recursively	`rm -r ~/new-lab`	Removes folder + contents

`rm -rf`	Force delete recursively	`rm -rf ~/os-lab`	Deletes without prompt
`grep`	Search text inside files	`grep "main" hello.txt`	Shows lines with `main`
`find`	Locate files	`find ~/ -name "hello.txt"`	Prints full path
`*` (globbing)	Match many filenames	`grep "int" *.txt`	Search in all `.txt` files
`>` (redirect)	Send output to file	`echo "OS" > hello.txt`	File content becomes `"OS"`
Absolute path	Full path from `/`	`/home/farhan/os-lab/hello.txt`	Always works globally
Relative path	Path from current dir	`./hello.txt` (inside `os-lab`)	Shorter reference
`chmod` (symbolic)	Change permissions	`chmod +x hello.txt`	Adds execute permission
`chmod` (numeric)	Set exact permissions	`chmod 644 hello.txt`	`rw-r--r--` (owner rw, others r)
Access control (in `ls -l`)	File rights (r=read, w=write, x=execute)	`-rw-r--r--`	Owner=rw, group=r, others=r
`-`, `d`, `l`	File types in `ls -l`	`-` file, `d` dir, `l` link	Example: `drwxr-xr-x new-lab`
Dotfiles	Hidden files start with `.`	`.bashrc`, `.gitignore`	Need `ls -a` to see

Exercise

Use the `man` command to find **at least 5 different usages** (options/flags) for each of the following commands:

- `cat`
- `ls`
- `cp`
- `touch`
- `rm`
- `mkdir`
- `pwd`
- `grep`

Lab report format

For each command, identify five different usages. For each usage, provide a brief description of its function and include a screenshot of the command's output.

Example:

Command: `ls -l`
Description: Displays files in long listing format (permissions, owner, size, date).
Screenshot: *(insert image)*